

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0703

Dooville 230 kV
March 17, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0703. The Interconnection Customer proposes a 828.3 MW natural gas combined cycle facility interconnecting at the Dooville 230 kV substation in the Indiana area. The request was received on November 16, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 299 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$106,594,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Blue Ravine-School Substation / Thomas Hill Station 230 kV Line
Contingency: Muskogee / Montpelier Electric Generating Sta 500 kV
Loading: 126.0% (Rating: 562 MVA)
DFAX: 16.9%
Pre-project loading: 101.2%
2. Facility: Thomas Hill Sta / Dooville 230 kV line
Contingency: Dooville / Montpelier Electric Generating Sta 230 kV
Loading: 119.5% (Rating: 674 MVA)
DFAX: 100.0%
Pre-project loading: 3.5%
3. Facility: Dooville / Montpelier Electric Generating Station Substation 230 kV
Contingency: Montpelier Electric Generating Station / Wabash Substation 500 kV
Loading: 112.2% (Rating: 908 MVA)
DFAX: 53.8%
Pre-project loading: 63.1%
4. Facility: W Melrose Substation / Timber Switch 230 kV
Contingency: Trego / Swan Valley 500 kV
Loading: 108.3% (Rating: 513 MVA)
DFAX: 6.5%
Pre-project loading: 97.7%
5. Facility: Muskogee / Timber Switch 230 kV
Contingency: Swan Vly / Trego 500 kV
Loading: 106.5% (Rating: 482 MVA)
DFAX: 6.6%
Pre-project loading: 95.1%

6. Facility: Stewart Mountain Dam / Ritter Park 345 kV line
Contingency: P.T.S / Medora 345 kV Line
Loading: 104.1% (Rating: 1,471 MVA)
DFAX: 7.1%
Pre-project loading: 100.1%
7. Facility: Simpson Stuart Tap / Fishkill Plains 345 kV L
Contingency: Medora / Hampton Rd 345 kV Line
Loading: 103.6% (Rating: 1,517 MVA)
DFAX: 6.1%
Pre-project loading: 100.3%
8. Facility: Burwell 500-345 kV Transformer Bank
Contingency: Allegheny Energy Units 3 4 & 5 345-1 kV TX
Loading: 103.1% (Rating: 1,004 MVA)
DFAX: 6.1%
Pre-project loading: 98.1%
9. Facility: Piney Crk Project / Hilltop 500 kV Line
Contingency: Caldwell / Piney Creek Project 345 kV
Loading: 101.5% (Rating: 2,515 MVA)
DFAX: 11.0%
Pre-project loading: 97.8%
10. Facility: Burwell / P.T.S 345 kV
Contingency: Allegheny Energy Units 3 4 & 5 345-1 kV TX
Loading: 100.9% (Rating: 1,484 MVA)
DFAX: 7.2%
Pre-project loading: 96.9%
11. Facility: Southeast Vicksburg / Armstrong 500 kV
Contingency: Hilltop / Piney Crk Project 500 kV
Loading: 100.1% (Rating: 1,898 MVA)
DFAX: 8.7%
Pre-project loading: 96.3%
12. Facility: Medora / Hampton Rd 345 kV
Contingency: Simpson Stuart Tap / Fishkill Plains 345 kV
Loading: 100.0% (Rating: 1,597 MVA)
DFAX: 7.0%
Pre-project loading: 96.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Blue Ravine-School Substation / Thomas Hill Station 230 kV Line	Rebuild line with higher-capacity conductor	\$35,226,000	25% (\$8,686,000)
Thomas Hill Sta / Dooville 230 kV line	Rebuild line with higher-capacity conductor	\$42,891,000	100% (\$42,891,000)
Dooville / Montpelier Electric Generating Station 230 kV	Rebuild line with higher-capacity conductor	\$32,212,000	57% (\$18,363,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Melrose Substation / Timber Switch 230 kV	Reconductor line	\$40,075,000	20% (\$7,851,000)
Muskogee / Timber Switch 230 kV	Reconductor line	\$6,614,000	20% (\$1,298,000)
Stewart Mountain Dam / Ritter Park 345 kV L	Reconductor line	\$6,465,000	8% (\$484,000)
Simpson Stuart Tap / Fishkill Plains 345 kV L	Reconductor line	\$15,933,000	7% (\$1,185,000)
Burwell 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$41,749,000	8% (\$3,137,000)
Piney Creek Project / Hilltop 500 kV	Reconductor line	\$253,945,000	8% (\$19,162,000)
P.T.S / Burwell 345 kV	Reconductor line	\$5,481,000	8% (\$411,000)
Southeast Vicksburg / Armstrong 500 kV	Reconductor line	\$24,094,000	7% (\$1,795,000)
Hampto Rd Substation / Medora Substation 345 kV Line	Reconductor line	\$17,820,000	8% (\$1,331,000)
	Total allocated to project		\$106,594,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.